ERGent

ERGent User Manual

Team ERG · Version 2+

A control-chart dashboard for 911 EMS dispatch in Ithaca, NY

Cornell Six Sigma Hackathon · Built October 17, 2026

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WELCOME

About us

> **This page is a template.** Every line in square brackets, like **[TO FILL IN: ...]**, is a placeholder. Team ERG: replace each one with your own words, then rebuild the manual (see the `README.md` in this folder). Delete this box when you are done.

Team ERG

[TO FILL IN: one short paragraph about the team. Who you are, what class and program you are in, and how you worked together during the Cornell Six Sigma Hackathon. Two to four sentences.]

Why we built ERGent

[TO FILL IN: the "why". Why did your team pick the emergency response prompt? What problem did you want to solve for the people who work in a 911 center? What do you hope a supervisor or a dispatcher will feel when they use ERGent? Three to five sentences.]

Meet the team

Nidhi Baderiya

Field	Your answer
Photo	[Photo placeholder: add a square photo, about 400 x 400 pixels]
Role on the team	[TO FILL IN: role on the team]
Program	[TO FILL IN: program, for example "Cornell M.Eng. Systems Engineering"]
Bio	[TO FILL IN: two-sentence bio]
What I built	[TO FILL IN: what I built]
Contact	[TO FILL IN: LinkedIn and/or email]

Zeba Haq

Field	Your answer
Photo	[Photo placeholder: add a square photo, about 400 x 400 pixels]
Role on the team	[TO FILL IN: role on the team]

Field	Your answer
Program	[TO FILL IN: program, for example "Cornell M.Eng. Systems Engineering"]
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Contact	[TO FILL IN: LinkedIn and/or email]

Dhruv Bhandari

Field	Your answer
Photo	[Photo placeholder: add a square photo, about 400 x 400 pixels]
Role on the team	[TO FILL IN: role on the team]
Program	[TO FILL IN: program, for example "Cornell M.Eng. Systems Engineering"]
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What I built	[TO FILL IN: what I built]
Contact	[TO FILL IN: LinkedIn and/or email]

Thanks

[TO FILL IN: thank-yous. For example your professor, mentors, classmates, and anyone who tested ERGent. Optional.]

Built with an AI coding agent, as the hackathon encourages. The team reviewed the work and can explain every number ERGent prints.

CHAPTER 1

Getting started

What ERGent is

ERGent is a dashboard for a city 911 ambulance (EMS) dispatch center. Here, the city is **Ithaca, NY**. It records the five times of every call: when the call arrived, when an ambulance was dispatched, when it left, when it reached the scene, and when it cleared. From those times it works out how long each part of a call took. It then shows a supervisor **which zone** is slow, **which part of the call** is slow, and **since when**. It flags zones that need attention, suggests a fix with its evidence, and later checks whether the fix worked. For the dispatcher on shift, it shows live clocks, so a late call is hard to miss.

ERGent only **flags and suggests**. It never dispatches an ambulance or changes staffing. People decide.

Why we built it this way

- A monthly average hides the slow calls. So ERGent also shows the 90th and 95th percentiles and uses control charts.
- Splitting each call into **stages** shows *why* a call was slow: dispatch, crew turnout, or travel.
- A chart for each **zone** shows *where* the problem is, and *since when*.
- A **before and after** check shows whether a fix really worked.

Who it is for

Role	When they use it	What they decide
Supervisor (the main user)	At the start of each shift, and whenever a flag appears	Which zone needs attention, which stage causes it, which fix to try, and later whether the fix worked
Dispatcher (the second user)	In real time, during a shift	Which open call is running late right now

How to start it

You need Python 3.12. The first time only, make a virtual environment and install the packages. Run these in the project folder:

```
python3.12 -m venv .venv
.venv/bin/pip install -r app/requirements.txt
```

Then start the app with one command:

```
cd app && ./testme.sh
```

Open **http://localhost:8000** in your browser. Stop the app with Ctrl+C in the terminal.

Live version: LIVE URL: (added after deploy). If your team has deployed ERGent, open that address instead. You do not need to log in.

> **Internet needed.** The Ithaca map and its tiles load from the internet. If the map area is blank, check your connection.

Choose a role

The start page has two buttons:

- **I'm a dispatcher** opens the live board and the History tab. Read chapter 2.
- **I'm a supervisor** opens the Charts and Actions tabs. Read chapter 3.

You can change role at any time. Click **Switch role** in the top right corner. Each browser tab keeps its own board, so two people can try ERGent at once without affecting each other.

The data is synthetic

All the data in ERGent is **invented**. It was made by a script with a fixed random seed, so it is the same every time. It is shaped like real field data, but it contains **no real patients, addresses or names**. Nothing here describes real Ithaca emergency calls.

There are three data sets in the Dataset menu (the supervisor view):

Data set	What it is for
Demo (42 days)	Has planted problems, so the tool has something to find. Zone D's crew turnout gets slow from Sep 15 and a fix is logged on Sep 29. Zone C slows in the weekday rush hour from Sep 20.
Baseline (28 days)	A stable city. A quiet dashboard is the right answer.
Messy file (3 days)	Has planted data errors: missing times, out-of-order times, a duplicate call ID. ERGent lists them and leaves them out.

The dispatcher's **Simulate a shift** button replays one day of the demo data.

The six zones

ERGent splits Ithaca into six zones. The outlines on the map are **illustrative**. They are not the official EMS districts.

Zone	Name
A	West Hill
B	Northside & Fall Creek
C	Cornell & Collegetown
D	West End & Southwest
E	Downtown & South Hill
F	East Hill & Belle Sherman

On the map, A, B and C run west to east along the north side. D, E and F run west to east along the south side. Zones that touch each other are neighbors. ERGent uses that when it suggests which zone should cover for another.

What the screens look like

The colors and layout can change between versions of ERGent. This manual describes screens by their **titles and controls**, so it still works if the look changes.

Targets used everywhere

Each call must meet these targets in **90% of calls**:

Part of the call	Priority 1	Priority 2	Priority 3
Dispatch (call received to dispatched)	64 s	64 s	64 s
Turnout (dispatched to en route)	60 s	60 s	60 s
Travel (en route to on scene)	8:00	14:00	19:00
Response (dispatched to on scene)	**9 min**	15 min	20 min

The priority 1 numbers come from NFPA 1710, a US fire and EMS standard. **The priority 2 and 3 targets are our own assumptions**, because the standard does not set them. A supervisor can change every target (see chapter 3).

CHAPTER 2

Dispatcher guide

This chapter is for the dispatcher on shift. It walks through every part of the dispatcher screen, in the same order as the numbers on the screen. Click **I'm a dispatcher** on the start page to begin.

The screen at a glance

- The **header** holds the simulation controls (Day, Speed, Start, **Simulate a shift**, **Reset**), a chip that shows the clock mode (such as `LIVE 1x` or `SIM 120x`), and **Switch role**.
- Two tabs sit under the header: **Live board** and **History**.
- **Live board** is for the calls happening right now. **History** looks back over the current shift and the one before it.

How to read a clock

Every open call has a running clock. Each clock has a target:

Clock	Target
Waiting (call received to dispatched)	1:04
Turnout (dispatched to en route)	1:00
Travel (en route to on scene)	8:00 for priority 1, 14:00 for priority 2, 19:00 for priority 3
Total (dispatched to on scene)	9:00 for priority 1, 15:00 for priority 2, 20:00 for priority 3

A clock gives you a **warning at 80% of its target**. It changes again when it is **past the target**. Past the target means the call is late in that stage. A supervisor can change the targets (chapter 3).

The top row: counts

Five counters show the workload: **Waiting**, **Dispatched**, **En route**, **On scene** and **Units available**. They update by themselves.

1. Log a call

Use this card to enter a call by hand. In a real center, the call-taker's system would do this for you.

1. Choose the **Zone**. The zone letters match the Ithaca map (A West Hill, B Northside & Fall Creek, C Cornell & Collegetown, D West End & Southwest, E Downtown & South Hill, F East Hill & Belle

Sherman).
2. Choose the **Priority**: **1: Life-threatening**, **2: Urgent** or **3: Non-urgent**.
3. Optionally choose the **Clock speed** (see "Clock speed" below). Leave it at 1x for real time.
4. Click **Add call**.

The call gets an ID like `LIVE-20261016-001`. Its **received time is stamped from the board clock**. You do not type any time.

2. Incoming calls

New calls wait here until you dispatch an ambulance. Each row shows the call ID, the zone, the priority, and a **Waiting** clock.

1. Look at the unit menu on the row. It lists only **available** ambulances. Ambulances from the call's own zone come first. Ambulances from other zones are marked with their home zone.
2. Pick the ambulance you want.
3. Click **Dispatch**.

If no ambulance is free, the row says **No unit free**. Wait for one to clear, or bring one back into service (see "Ambulances").

If the call has waited longer than its 1:04 target, ERGent asks for a delay reason first (see "The delay-reason dialog").

3. Dispatched

The call is assigned and the crew must get moving. The row shows the unit and a **Turnout** clock. When the crew leaves, click **En route**. If turnout ran past 1:00, ERGent asks why.

4. En route

The ambulance is on the road. The row shows two clocks: **Travel** and **Total**. When the crew arrives, click **On scene**. If travel ran past its target, or the total response ran past its target, ERGent asks why.

If the ambulance comes from another zone, the row says so, for example "(from zone A)".

Ambulances

This card shows all 12 ambulances in four columns. Each column has a count:

- **Available**: free to be dispatched.
- **Responding**: dispatched or en route. The chip says which.

- **On scene**: at the scene, not yet cleared.
- **Out of service**: not usable.

Each ambulance shows its ID and home zone, and one button.

- **Out of service** takes an ambulance out of use, for example after a breakdown. If it was rolling to a call you logged, that call is flagged **Breakdown**. ERGent records "ambulance breakdown" as the reason when the call reaches the scene, so you do not have to.
- **Back in service** returns an ambulance to the **Available** column.

Zone map

The map shows Ithaca with the six zones. The outlines are **illustrative**, not official EMS districts.

- **Hover** over a zone to see its name, its status, how many calls are open there and how many ambulances are available.
- **Click** a zone to pick it in **Log a call**.
- The status comes from the clocks of that zone's open calls. A zone shows **On track** when all its clocks are comfortable, **Watch** when a clock is above 80% of its target, and **Past target** when a clock is late.
- The map needs an internet connection to draw.

5. Resolved

A call moves here **when it reaches the scene**. "On scene" counts as resolved. The list shows the latest 60 calls and scrolls.

Each row shows the unit, the time, the delay reason (if any), and the **final response time**. The response time is marked as late when it passed its target.

- A call that you logged by hand, and that is still at the scene, has a **Clear** button. Click it, then confirm "Clear call ...?". The row changes to **Cleared** and the ambulance becomes available.
- Replayed calls (from **Simulate a shift**) show **At the scene** and **Replay**. They are read-only.

The delay-reason dialog

When you press a button that ends a stage that ran **past its target**, ERGent will not accept the press until you say why. A dialog titled **Why was this late?** opens. It tells you how long the stage took and what the target was.

1. Choose one reason from the list:
    - Traffic or road closure
    - Ambulance breakdown

    - Unit from another zone
    - Unclear address
    - Weather
    - Crew not ready
2. Click **Save reason and continue**.

The time you pressed the button is kept, not the time you chose the reason. If you click **Cancel**, the dialog closes and the press is not recorded.

The reason is saved with the call. The supervisor's suggested actions use these reasons.

The rules that protect you (mistake-proofing)

ERGent is built so that wrong entries are hard to make:

- Only the **next valid button** works: **Dispatch**, then **En route**, then **On scene**, then **Clear**. Any other press is refused with a message.
- You can only dispatch an **available** ambulance.
- A **late** stage needs a **reason**. The reason is one of six fixed codes, so it can be counted.
- **Clear** asks for confirmation first.
- Times come from the clock. You never type a time.

Clock speed

The **Clock speed** menu on the **1 Log a call** card sets how fast the board clock runs when **no simulation is running**. The choices are **1x**, **10x**, **30x** and **60x**.

- At **1x** the board clock is real time.
- At **60x**, one real second is one minute on the board. This is good for practice, because clocks turn late in seconds.
- Changing the speed **continues from the current board time**. The clock never jumps.
- At a high speed the board clock runs ahead of real time. The shift label in **History** follows the board clock.

The header chip shows the speed, for example `LIVE 60x`.

Simulate a shift

The header controls replay one day of the demo data on a fast clock. This is the best way to learn ERGent.

1. Choose the **Day** (the default is Sep 20).

2. Choose the **Speed**: **30x**, **60x** or **120x**.
3. Choose where to **Start**: **00:00**, **06:00**, **14:00** or **22:00**.
4. Click **Simulate a shift**.

The chip changes to `SIM 120x` (or the speed you chose). Calls appear and move through cards 2, 3, 4 and 5 at the times in the data. The button becomes **Pause**. Then it becomes **Resume**, and **Run again** when the day ends.

Good to know:

- Replayed calls are **read-only**. You can still add your own calls, and they use the same clock.
- Calls that were already cleared before the start time are left out.
- You can change the **Speed** while it runs.
- To change the day or the start time, click **Reset** first.

Reset

Reset clears everything back to zero:

- All replayed and hand-logged calls disappear.
- **Waiting**, **Dispatched**, **En route** and **On scene** all show 0.
- All 12 ambulances are **Available** again.
- **Resolved** is empty.
- The clock goes back to real time, at the clock speed you last chose.

Download calls

Download calls (CSV) saves this session's calls as a spreadsheet file. The columns are the same as the data files in the project, so you can open it in Excel or load it back into ERGent's tools.

History tab

History looks back over the **current shift and the one before it**. A shift is eight hours: **day** 06:00 to 14:00, **evening** 14:00 to 22:00, **night** 22:00 to 06:00. History follows the board clock, so during a simulation it follows the simulated time. It refreshes about every 5 seconds while you look at it.

From top to bottom:

1. **The shift label.** For example: "Current: day shift 06:00-14:00, Sep 16 · Previous: night shift 22:00-06:00, Sep 15".
2. **Four numbers: Calls**, **Mean (min)**, **Standard deviation (min)** and **On time**. They cover all priorities and only calls that have reached the scene. "On time" means within each call's own target. The mean is

the average response time. The standard deviation shows how spread out the times are.

3. **Response time by zone** (the chart). Use the **Priority** menu to switch between **Priority 1: Life-threatening**, **Priority 2: Urgent** and **Priority 3: Non-urgent**. Each zone has a box that shows where most of its calls fall, and a diamond for its average. A line shows the city's center, short dashed lines show each zone's limits, and a line shows the target.
   - A **red diamond** is a zone that is **slower than the whole city**. **Hover over it** to see why. For example: "Zone D is slower than the city: 9.3 min typical vs. 6.3 min. Main cause: crew turnout (100% of minutes over target). Top reason: crew not ready (2 of 2 late calls)."
   - A zone marked **faster than the city** is shown as "check the data". Be careful: very fast times can be a data mistake.
   - Under the chart, a note names the slower zones, or says "No zone is slower than the city as a whole."
   - The number under each zone letter (such as `n=5`) is how many calls it has. A zone with few calls has wide limits, so be careful with it.
4. **Suggested actions** (on the right). ERGent's live tips for you. See below.

Right after you start ERGent in real time, History is empty, because the demo data is from the past. It fills up when you log calls or run **Simulate a shift**.

Suggested actions (tips T1 to T6)

Each tip has a short name, a zone tag, a plain sentence, and the evidence in small type. A note reminds you: "For the dispatcher's judgment; the tool decides nothing on its own."

Tip	When it appears	What it suggests
T1 Dispatch now	A call on the board has waited longer than 64 s for dispatch	Dispatch it now. It names the nearest free ambulance: one from the call's own zone first, then one from a neighboring zone. If none is free, it says so.
T2 Rebalance	A zone has no free ambulance and a neighboring zone has two or more	Move one ambulance to stand by in the empty zone.
T3 Cover	An ambulance is out of service	Ask the neighboring zones to cover that zone's calls.
T4 Route	A zone's typical travel time is at least 1.25 times the city's (needs 3 or more calls in the two shifts)	Check the route. If two or more late calls share a reason, the tip names it: reroute around traffic or road works, or keep an ambulance posted in that zone.
T5 Station alert	A zone's typical crew turnout is over 60 s (needs 3 or more calls)	Alert that zone's station: crews are slow to get rolling.

Tip	When it appears	What it suggests
T6 Dispatcher load	Fewer than 9 in 10 calls were dispatched within 64 s over the two shifts (the 90th percentile of dispatch time is over 64 s; needs 5 or more calls)	Ask the supervisor for a second dispatcher.

If there is nothing to suggest, the card says "No suggestions right now."

What to do when...

Situation	What to do
A call has waited too long	Pick an ambulance in **2 Incoming calls** and click **Dispatch**. The **T1 Dispatch now** tip in History names the nearest free one.
The row says **No unit free**	Wait for an ambulance to clear. If one is out of service and ready again, click **Back in service**.
An ambulance breaks down	Click **Out of service** on it. Its call is flagged **Breakdown**, and the reason is recorded for you.
A button you expect is missing	Only the next valid button is shown. Check which card the call is in.
ERGent asks "Why was this late?"	Choose the real reason and click **Save reason and continue**. Your press time is kept.
You pressed a button by mistake before the dialog	Click **Cancel** in the dialog. The press is not recorded.
You want to practice	Click **Simulate a shift**, or add calls and set **Clock speed** to 60x.
You want to start again	Click **Reset**.
You want to see how the shift is going	Open **History**.
A zone is red in History	Hover over the red diamond to read why. Look at the **Suggested actions** next to it.
A zone has no free ambulance	Look for **T2 Rebalance** in History.
You want to keep a record	Click **Download calls (CSV)**.
The map is blank	Check your internet connection. The map loads from the internet.

CHAPTER 3

Supervisor guide

This chapter is for the EMS operations supervisor. The supervisor view helps you answer four questions, in this order:

1. **Which zone** needs attention?
2. **Which part of the call** (dispatch, crew turnout or travel) is slow?
3. **What should we try?**
4. **Did the fix work?**

Click **I'm a supervisor** on the start page. You get two tabs: **Charts** and **Actions**. Charts answers questions 1 and 2. Actions answers questions 3 and 4.

Every card has one plain title. Under most cards, a **More info** toggle opens a short explanation of the card in two or three sentences. Use it whenever you want to know what you are looking at.

The Charts tab

The control bar

The control bar at the top of Charts sets what **both tabs** show.

Control	Choices	What it does
Chart	**Daily chart**, **Every call**	Which control chart to show (one at a time)
Zone	Zone A to F, with its name	Which zone the control chart shows. You can also click a zone on the map.
Priority	**1: Life-threatening**, **2: Urgent**, **3: Non-urgent**	Which calls to look at. Each priority has its own target.
Dataset	Demo, Baseline, Messy file	Which data to look at (see chapter 1)
As of	A date in the data	The "today" for every card. Later calls are ignored. It moves to the last day of the data when you change the dataset.
Chart period	Last 14, 28 or 42 days	How much history every card uses. **It also sets the control limits** (see below).
Targets	A button	Opens the Targets dialog (see the end of this chapter)

> **Tip.** To follow the demo story, choose the **Demo** data set and set **As of** to **2026-09-28**. That is the day before the fix. Then set it to **2026-10-12** to see the result.

Control chart

The control chart is the main picture. It shows whether a zone is behaving the way it normally does. Choose **Daily chart** or **Every call**.

Daily chart

The Daily chart has two panels.

Top panel: "Zone X: daily average response."

- Each **dot** is **one day**. Its height is that day's typical response time in minutes. ("Typical" is a geometric average. It is a fair average for times with a long slow tail.)
- The **center line** is the zone's normal level. It is the average of the daily dots in the chart period.
- The two **dashed lines** are the **control limits**. They are the upper control limit (**UCL**) and the lower control limit (**LCL**). They mark the normal range. They are the same for every day.
- A **red dot** is a day that broke a flag rule (see below). Something probably changed.
- A **hollow ring** is a "check the data" day: unusually *fast*. Check the records before you trust it.
- The **amber line** is the **target** (the specification limit). It is a requirement from the standard. It is not part of the control limits.
- A **green vertical line** marks a logged action (a fix).

Bottom panel: "Daily spread." It shows how spread out that day's calls were. A red dot means some calls that day were much slower than others.

What makes a dot red? One of these rules fired on the slow side:

Rule	A red dot appears when...
R1	one day is far outside the normal range (beyond 3 sigma)
R2	2 of the last 3 days are well above normal (beyond 2 sigma)
R3	4 of the last 5 days are above normal (beyond 1 sigma)
R4	8 days in a row are above the center line
R5	6 days in a row are getting slower

A day on the *fast* side only says "check the data". It is not an alarm.

> **Important: limits come from the chart period.** The control limits are worked out from the days in the **chart period** you selected. That is simple, and it keeps the chart, the map, the suggestions and the capability card all in agreement. But there is a trade-off. If a problem lasts for most of the period, it **widens the limits** and can **hide itself**. Example: in the demo data, as of Sep 28 with a 28-day period, Zone D's daily chart shows **no red dots**, because D's slowdown (from Sep 15) is inside the window. The other cards still show the problem: the **Zone map** is amber for D, **Delay by stage** blames crew turnout, and **Capability** says D is below the standard. **Always read the daily chart together with those cards.**

Every call

Every call shows each call, in the order the calls came in. The height is the response time in minutes. The scale is a log scale, so the slow tail does not squash the rest.

- The **center line** is the typical call.
- The dashed **UCL** and **LCL** are the normal range for a *single* call.
- A **red dot** is a call **above the UCL**: an unusual delay worth looking at one by one.
- The **amber line** is the target. Calls above it missed the target.

Statistics

The **Statistics** card sums up the chart period for **all zones**, for the priority you chose. The line under the title shows the dates and the number of calls.

Number	What it means
Average	The mean response time in minutes
Standard deviation	How spread out the response times are. A big number means less predictable service.
On time	The share of calls within the target. The goal is **at least 90%**.
Zones flagged	How many zones are amber or red on the map, out of 6

Zone map

The **Zone map** shows Ithaca in six zones. The outlines are **illustrative**, not the official EMS districts. Each zone is colored by its status over the **last 7 days**:

Status	Meaning
On track	Nothing to worry about
Worth a look (amber)	One warning sign, either a real change on the chart or a 90th percentile over the target
Act now (red)	Both warning signs together

Status	Meaning
Not enough data	Too few calls to judge

- **Hover** over a zone to see its status and the reasons, such as "90th percentile 10.2 min vs. 9 min target" and "turnout causes 99% of minutes over target".
- **Click** a zone to select it. The **Zone** menu and the control chart switch to that zone.
- A zone that is slipping back after a logged fix is at least amber.
- A single weak sign (for example, one rule fired once) does not turn a zone amber. It is mentioned in the zone's reasons: "one weak signal ... keep watching". This keeps false alarms rare.

Data quality

This card appears **only when some calls were left out** of the statistics. It lists each call and why: a missing time, times in the wrong order, a duplicate call ID, a zero response time, or an unknown zone or priority. These calls are never counted, so one bad record cannot distort a chart. The **Messy file** data set shows this card.

Delay by stage

A call has three stages: **dispatch processing**, **crew turnout** and **travel**. This card shows **which stage causes the minutes over target** in each zone. The line under the title shows the dates and the priority.

- Each zone has one bar. The bar is split into three parts. Each part is that stage's **share of the zone's minutes over target**. The bar adds up to 100%.
- The last bar, **City**, is all zones together.
- **Hover** over a part of a bar to see its share, the total minutes over target, and the median time of that stage.

How to read it: the biggest part of a zone's bar is the stage to fix first. In the demo data, Zone D's bar is almost all crew turnout. Most other zones are mostly travel.

Zone comparison

This card answers "**do the zones really differ?**" in a sentence. For example: "Yes. Zone D is typically 6.0 min against 4.9 min in Zone A; a gap this big is very unlikely to be chance." If the gaps are small, it says "No clear difference: the gaps between zones could be chance."

Open **More info** for the test result (the F value and p-value) and a probability plot. See chapter 4 if you want the details.

Capability

The **Capability** card asks: "**Can each zone meet the target, reliably?**" It gives one plain verdict for each zone.

Verdict	What it means
Very reliable	99 or more calls in 100 arrive within the target
Reliable	95 to 98 calls in 100 arrive within the target
Just meets the standard	90 to 94 in 100. There is little margin.
Below the standard	Under 90 in 100
Can't judge yet	The zone's times changed during the period, so a score would mislead. Fix the cause first.
Not enough calls	Fewer than 30 calls in the period

Open **More info** for the Ppk and Cpk numbers behind the verdicts. The standard asks for at least 90 calls in 100 to be on time.

The Actions tab

Actions answers "**what should we try, and did it work?**" A line at the top reminds you of the data set, priority and date. Change them on the **Charts** tab.

Slip-back alerts

This card appears **only when a zone is slipping back** after a logged action. It names the zone and the warning sign. A zone that slips back is at least amber on the map.

Suggested actions

This card lists fixes ERGent suggests. Each one has:

- a **tag** for the kind of fix: **Crew readiness**, **Coverage**, **Routes**, **Dispatcher cover**, **Maintenance** or **Address checks**;
- a **zone tag** (or **Citywide**);
- the **suggestion**, in plain words;
- the **evidence**, so you can see why;
- an **estimate**, such as "on time 78% → 97%". This is a best case: what the on-time share would be if that stage were as fast as the city's typical call.

Every suggestion is "for review by qualified staff". ERGent decides nothing. See chapter 4 for the rules behind S1 to S6.

To log a suggestion as an action:

1. Click **Log this action** on the suggestion.

2. A dialog **Log an action** opens, already filled in: the **Zone**, the **Stage**, a **Start date** (the day after the as-of date), **What was done** and the **Owner**. Edit anything you like.
3. Click **Save action**.

If that same action is already in the log, ERGent says "Already in the log as A1" and does not add it twice. Once logged, the suggestion shows its status, for example "Action logged on 2026-09-29".

Actions taken

This is the action log. Each row has an ID (A1, A2...), the zone, the stage, the date, what was done and who owns it. An action dated after the as-of date is tagged **After the as-of date**.

Click **Log an action** to add one by hand, without a suggestion. The demo data already holds one action: **A1**, a night crew readiness review at Zone D's station, dated 2026-09-29. Actions are only kept for the length of your session.

Before and after

This card shows the **daily chart** for the action's zone, with the action marked by a green line. You can see whether the days after the fix stay in the normal range.

Result

The **Result** card tells you whether the fix worked.

1. Choose the action in the menu (the latest action on or before the as-of date is chosen for you).
2. Read the **verdict chip**:
   - **Improved**: the drop is too large to be chance.
   - **Got worse**: the times got slower, and too much to be chance.
   - **No clear change yet**: the change could be chance.
   - **Not enough data**: there are not yet enough calls after the action. For example, if the as-of date is before the action.
3. Read the table. It compares the **14 days before** the action with the **days after it** (up to 14):

Row	Before	After
Mean	average response time	average response time
Standard deviation	spread of the times	spread of the times
On time	share within the target	share within the target
Calls	number of calls	number of calls

Open **More info** for the test numbers and to see whether the new level is **steady**: whether the days after the action stay inside the normal range.

> **Watch out.** A better time after a fix does not *prove* the fix caused it. Other things may have changed. The check shows an apparent change. Staff judge the cause.

The Targets dialog

Click **Targets** in the control bar to see and change the targets. The dialog has five numbers:

Field	Default
Priority 1 target (min)	9
Priority 2 target (min)	15
Priority 3 target (min)	20
Processing target (s)	64
Turnout target (s)	60

Click **Save** to apply them, or **Cancel** to close the dialog. The new targets change every card.

The defaults come from NFPA 1710, a US standard, for priority 1: 64 seconds to dispatch, 60 seconds for the crew to get moving, and 8 minutes to travel, which makes 9 minutes from dispatch to arrival. **The priority 2 and 3 targets are our own assumptions.** The travel target is always the response target minus the 60-second turnout target.

CHAPTER 4

Statistics explained

Every number ERGent shows comes from **one file of plain, tested functions** (`app/qc.py`). The screens only draw those results. This chapter explains each statistic **in plain words first**, then gives the formula. Each one has four parts:

- **What it tells you**
- **How to read it**
- **The formula**
- **Watch out for**

You do not need this chapter to use ERGent. It is here so you can **explain every number**.

Where each statistic appears

Statistic	Where you see it
Stage times and data checks	Behind everything. Data quality card.
Mean, standard deviation, percentiles, % on time	Statistics card, History numbers, Result table
Daily $\bar{X}$-S chart	Charts tab, Control chart (Daily chart)
Every-call chart	Charts tab, Control chart (Every call)
Flag rules and zone health	Red dots, Zone map (both views)
Zones-as-subgroups box chart	Dispatcher History, Response time by zone
Delay by stage	Charts tab, Delay by stage
Capability	Charts tab, Capability
ANOVA	Charts tab, Zone comparison
Welch's t-test and before/after	Actions tab, Result
Impact estimate and suggestion rules S1 to S6	Actions tab, Suggested actions
Dispatcher tips T1 to T6	Dispatcher History, Suggested actions
Slip-back	Actions tab, Slip-back alerts

Three ideas to know first

1. Times are in seconds inside the data and in minutes on screen. A response of 540 seconds is 9 minutes.

2. Control charts and capability use the logarithm of the time. Response times are *skewed*: most calls are quick, and a few are very slow. The log (written `ln`) pulls in that slow tail. After the math is done, ERGent turns the results back into minutes (with `exp`). So a "typical" time on a chart is a *geometric* average: `exp(mean of ln minutes)`.

3. Control limits are not the target. A **control limit** (UCL and LCL) comes from the process itself. It says what is *normal for this zone*. A **specification limit** (the 9-minute target) comes from a standard. It says what is *required*. A zone can be inside its control limits and still miss the target, and the other way round. ERGent draws the target as a separate line, so you never confuse the two.

1. Stage times and data checks

What it tells you. How long each part of a call took, so you can say *why* a call was slow.

How to read it. A call has three stages. The prompt's metric, **response**, is turnout plus travel.

The formula.

```
processing = dispatched  − call_received     (dispatch: the call waits for an ambulance)
turnout    = en_route    − dispatched         (the crew gets moving)
travel     = on_scene    − en_route           (the drive)
response   = on_scene    − dispatched         (= turnout + travel)
```

A call is **late** when its response is greater than its priority's target. A **stage** is late when it is greater than that stage's target. The travel target is the response target minus the turnout target: for priority 1, 540 – 60 = 480 seconds.

Data checks. A call is **left out of every statistic** and listed, with its reason, if:

1. its call ID is a **duplicate** (the repeat is left out; the first is kept);
2. a needed time is **missing** (call received, dispatched, en route or on scene);
3. a time is **earlier than the one before it**;
4. its response time is **zero**;
5. its **zone or priority is unknown**.

Watch out for. Processing happens *before* dispatch, so it sits outside the prompt's metric. ERGent still measures it, because a delay there also hurts the patient. Also, a crew might press "on scene" a little after arriving, so the times are only as good as the pressing habit.

2. Mean, standard deviation, percentiles and % on time

What it tells you. The **mean** is the average. The **standard deviation (SD)** says how spread out the times are. The **median** is the middle call. The **90th percentile** is the time that 9 in 10 calls beat. The **95th percentile** is the time that 19 in 20 calls beat. **On time** is the share of calls within the target.

How to read it. Look at the percentiles, not just the mean. The danger in emergency response is the slow tail, and the mean hides it. The standard says 90% of calls should be on time. That is the same as saying the 90th percentile should be at or under the target.

The formula.

```
mean        = (x1 + x2 + ... + xn) / n
SD          = sqrt( sum of (x − mean)^2 / (n − 1) )          (sample SD, n − 1)
on time (%) = 100 x (number of calls with x <= target) / n
percentile  : sort the values; position h = (n − 1) x p, counting from 0
              result = x[floor h] + (h − floor h) x ( x[floor h + 1] − x[floor h] )
```

This is the "type 7" percentile. It is the same as R's default and Excel's `PERCENTILE.INC`.

Worked example. Ten response times in minutes: 4, 5, 6, 7, 8, 9, 10, 11, 12, 20, with a 9-minute target.

Number	Value
Mean	9.2
Median	8.5
90th percentile	12.8 (h = 9 × 0.9 = 8.1, so 12 + 0.1 × (20 – 12))
95th percentile	16.4
On time	6 of 10 = 60%

The mean (9.2) looks close to the 9-minute target. But only 60% of the calls were on time, and one patient waited 20 minutes. That is why ERGent does not rely on the mean.

Watch out for. With one call, the percentiles equal that value and the SD cannot be worked out. With no calls, ERGent shows a dash.

3. Checking that logs behave (normality)

What it tells you. Whether the log of the response times is close to a bell curve. Several tests assume it is.

How to read it. Open **More info** on the **Zone comparison** card. A probability plot shows the points. If they sit close to a straight line, the log times look normal. The **Shapiro-Wilk** test gives a number `p`. A `p` above

0.05 means "no sign of a problem".

The formula. The Shapiro-Wilk W statistic and its p-value (from the SciPy library), on `ln(minutes)`.

Watch out for. With hundreds of calls, even a tiny bend can give a small `p`. Judge by the plot as well. ERGent needs at least 3 values that are not all the same.

4. The daily X̄-S chart

What it tells you. Whether a zone is behaving the way it normally does. Each dot is **one day**. A red dot is a day when the zone was slower than its own normal range.

How to read it. The top panel (X̄) shows each day's typical response time. The center line is the normal level. The dashed lines are the UCL and LCL. The bottom panel (S) shows how spread out the day's calls were. A red dot means something changed. Open **Delay by stage** to see which stage.

The formula. The subgroup is one zone-day. Work on `ln(minutes)`:

```
daily point     = mean of ln(minutes) for that day's calls
center          = mean of the daily points in the chart period
sigma_xbar      = standard deviation of the daily points in the chart period (n - 1)
UCL, LCL        = center +/- 3 x sigma_xbar         (the same for every day)

S chart:
s_bar           = mean of the daily standard deviations (days with 2 or more calls)
n_bar           = average number of calls per day in the period (rounded, at least 2)
c4(n)           = sqrt( 2 / (n - 1) ) x Gamma(n / 2) / Gamma( (n - 1) / 2 )
B4 = 1 + 3 x sqrt(1 - c4^2) / c4        B3 = 1 - 3 x sqrt(1 - c4^2) / c4   (never below 0)
S limits        = B4 x s_bar (upper) and B3 x s_bar (lower)
```

Here are some values of the constants (they match the standard tables):

Calls per day (n̄)	c4	B3	B4
2	0.7979	0	3.267
3	0.8862	0	2.568
5	0.9400	0	2.089
7	0.9594	0.118	1.882
10	0.9727	0.284	1.716

The limits are then shown back in minutes (`exp`). The chart is judged with the **flag rules** in section 6.

Watch out for: the chart period sets the limits. In ERGent, the limits come from the **chart period you select** (14, 28 or 42 days). The team chose this on purpose. It is simple, and it keeps the chart, the zone map, the suggestions and the capability card in agreement. But there is a cost. **If a problem lasts for most of the period, it widens the limits and can hide itself.**

For example, in the demo data, Zone D's crew turnout got slow on Sep 15. As of Sep 28 with a 28-day period, the period includes both normal days and slow days. The limits stretch to about 3.3 to 10.5 minutes, and **no day breaks them**. Zone D's chart shows **no red dots**. If the limits were taken from the first 14 days instead (before the problem), the same chart would flag Zone D on 11 days.

So: **read the daily chart together with the Zone map, Delay by stage and Capability.** In this example, the Zone map is amber for D (its 90th percentile is 10.2 min against a 9-min target), Delay by stage says 92% of its overrun is crew turnout, and Capability says D is below the standard.

Also watch out for days with very few calls. Their dots are noisy, and a one-call day has no spread, so it has no point on the S chart.

5. The every-call chart (individuals chart)

What it tells you. Whether a *single call* was unusually slow. It looks at each call, in the order the calls came in.

How to read it. Each dot is one call. The dashed lines (UCL and LCL) are the normal range for a single call. A red dot above the UCL is an unusual delay worth checking one by one: a breakdown, a wrong address, a closed road. The amber line is the target. The scale is a log scale.

The formula. This is the course's I-MR (individuals and moving range) method, on `ln(minutes)`:

```
moving range   MR_i = abs( x_i - x_(i-1) )           (jump from one call to the next)
MR_bar         = mean of the moving ranges
center         = mean of ln(minutes)
sigma          = MR_bar / 1.128                        (1.128 is the constant d2 for
pairs)
UCL, LCL       = center +/- 3 x sigma
MR upper limit = 3.267 x MR_bar                        (computed, not drawn)
```

Worked example. Five values 10, 12, 11, 15, 12. The moving ranges are 2, 1, 4, 3, so $\bar{MR}$ = 2.5. The center is 12 and σ = 2.5 / 1.128 = 2.216. The limits are 5.351 and 18.649.

Watch out for. It needs at least 2 calls. If every call is the same, there is no spread, so there are no limits ("no variation"). The calls are assumed to be roughly independent, which is stated but not tested. As on the daily chart, the limits come from the calls in the chart period.

6. The flag rules, and what makes a zone amber or red

What it tells you. The rules decide which dots on the daily chart turn red. The stricter **zone health** rule decides whether a zone on the map is on track, worth a look, or act now.

How to read it. A dot **above** the center line is on the **slow side**. That is an **alarm**. A dot **below** the center line is on the **fast side**. That only says "check the data".

The formula. Let `z = (daily point − center) / sigma_xbar`. So "1 sigma" means one `sigma_xbar`. The rules are:

Rule	A flag fires at a day when...
R1	that day's point is beyond 3 sigma
R2	2 of the last 3 points are beyond 2 sigma, on the same side
R3	4 of the last 5 points are beyond 1 sigma, on the same side
R4	8 points in a row are on the same side of the center line
R5	6 points in a row are steadily rising (or falling)

R1 to R4 are the **Western Electric rules**. R5 is a trend rule. Together they catch both sudden jumps and slow drifts.

Zone health. On the map, each zone looks at the **last 7 days**. First, ERGent decides if there is a **chart signal**:

```
chart signal = rule R1 fired in the last 7 days,
               OR 2 or more DIFFERENT slow-side rules fired in the last 7 days

red    ("Act now")       = chart signal AND the 90th percentile of the last 7 days is over
the target
amber  ("Worth a look")  = exactly one of those two
green  ("On track")      = neither
unknown                  = not enough data to set limits
```

A zone that is **slipping back** after a logged fix (section 18) is at least amber.

Why stricter? One weak rule on its own is drawn on the chart but does not count. It appears in the zone's reasons as "one weak signal ... keep watching". The team tested this. On pure-noise simulations, a stable zone-week shows amber by chance about **2.5%** of the time with this rule, against about **7.7%** if any single rule counted.

Watch out for. Even so, **some false alarms are expected**. Amber means "worth a look", not "something is broken". Red means act. A zone can also be amber only because its 90th percentile is just over the target, with no change on the chart.

7. Zones as subgroups (the dispatcher's History chart)

What it tells you. Which zones are slower than the city as a whole, in the current and previous shift.

How to read it. Each zone is drawn as a **box plot** (the box holds the middle half of its calls) with a **diamond** for its average. The grey line is the city's center. The short dashed lines are that zone's limits. A **red diamond** is a zone beyond its upper limit: slower than the city. A hollow ring is a zone beyond its lower limit: faster, so check the data. **Hover** over a red diamond to read why.

The formula. Each zone is **one subgroup** of `ln(minutes)` (calls that reached the scene, for the priority you chose):

```
center      = mean of the six zone averages
sigma_short = sqrt( mean of the zone variances )         (zones with 2 or more calls)
limits      = center +/- 3 x sigma_short / sqrt(n_zone)  (wider for zones with few calls)
z           = (zone average - center) / ( sigma_short / sqrt(n_zone) )
flag        = "above" if z > 3, "below" if z < -3
```

Zones have **no time order**, so only the "beyond the limit" rule is used. This is like an *analysis of means*.

The hover text for a red zone says: its typical time against the city's, the stage causing the most minutes over target, and the most common delay reason among its late calls.

Watch out for. A zone with few calls has wide limits, so it is hard to flag. A zone with no calls shows nothing. The two shifts together can be a small sample early in a shift.

8. Delay by stage

What it tells you. Which stage (dispatch, crew turnout or travel) causes the minutes over target.

How to read it. For each zone, the three parts of the bar add up to 100%. The biggest part is the stage to fix first. **Hover** over a part to see its share, the minutes over target, and the stage's median time.

The formula.

```
stage excess = max( 0, stage time - stage target )         for each call, in minutes
share        = a stage's total excess / the sum of the three stages' total excess
```

Watch out for. A share is only about **minutes over target**. A stage that stays under target has no share, even if it is slow compared with other stages. If nothing is over target, all shares are 0.

9. Capability

What it tells you. Whether a zone can meet the target **reliably**, not just on average.

How to read it. ERGent turns the result into a plain verdict.

Verdict	Expected calls on time (rounded)
Very reliable	99 in 100 or more
Reliable	95 to 98 in 100
Just meets the standard	90 to 94 in 100
Below the standard	under 90 in 100
Can't judge yet	The zone is not stable (see below)
Not enough calls	Fewer than 30 calls

The formula. The target is an **upper limit** only (a response cannot be "too fast"), so ERGent uses the **upper side** of the capability indices, on `ln(minutes)`:

```
USL        = ln( target in minutes )
Ppk        = ( USL − mean of ln x ) / ( 3 x SD of ln x )             uses the overall
spread (sigma_total)
Cpk        = ( USL − mean of ln x ) / ( 3 x sigma_short )             uses the spread
within each day
sigma_short = sqrt( mean of the daily variances )
expected on time (%) = 100 x Phi( 3 x Ppk )                           Phi = the standard
normal curve
```

Some handy values:

Ppk	Expected on time
0.33	83.9%
0.43	90.1% (the standard's 90% line)
0.50	93.3%
0.67	97.8%

Ppk	Expected on time
0.80	99.2%
1.00	99.9%

The verdict is judged on the **rounded** percentage, so the label always agrees with the number in the text.

Stable only. A capability score only makes sense for a **stable** zone. ERGent shows a score only when the zone's daily chart has **no slow-side flag in the chart period** *and* the zone has **at least 30 calls**. Otherwise it says "Can't judge yet" or "Not enough calls".

Watch out for. Cpk and Ppk differ when a zone drifts between days. Ppk includes the drift; Cpk does not. The numbers sit behind **More info**. The score assumes the log times are roughly normal. Because the daily chart's limits come from the chart period, a long problem can hide on the chart (section 4). In that case capability may still say "Below the standard", as it does for Zone D in the demo.

10. Zone comparison (ANOVA)

What it tells you. Whether the zones **really differ**, or whether the gaps could be chance.

How to read it. ERGent answers in a sentence: "Yes. Zone D is typically 6.0 min against 4.9 min in Zone A ..." or "No clear difference ...". The typical times are geometric averages.

The formula. A one-way **analysis of variance** (ANOVA) on `ln(minutes)` of each zone's calls:

```
SSB = sum over zones of  n_zone x ( zone mean − grand mean )^2      (between zones)
SSW = sum over zones of  sum of ( x − zone mean )^2                   (within zones)
F   = ( SSB / (k − 1) ) / ( SSW / (N − k) )                           k zones, N calls
p   = the chance of an F this big if the zones were really the same  (from the F
distribution)
```

ERGent says "Yes" when `p < 0.05`.

Worked example. Zone A = [1, 2, 3] and zone B = [4, 5, 6]. SSB = 13.5 and SSW = 4, so F = 13.5 / (4 / 4) = 13.5 with 1 and 4 degrees of freedom.

Watch out for. A zone with fewer than 2 calls is left out. If every call in each zone is identical, the test cannot run. ANOVA says *that* zones differ, not *which* one. Look at the typical times for that. A "Yes" with hundreds of calls can come from a small gap, so check that the gap is large enough to matter.

11. Welch's t-test and the before/after check

What it tells you. Whether response times really dropped after an action.

How to read it. The **Result** card shows a verdict: **Improved**, **Got worse**, **No clear change yet** or **Not enough data**.

The formula. ERGent compares the **14 days before** the action date with the **days from the action date on** (up to 14), using **Welch's one-sided t-test** on `ln(minutes)`. Welch's test does not assume both groups have the same spread.

```
t  = ( mean_before − mean_after ) / sqrt( s_before^2 / n_before + s_after^2 / n_after )
df = Welch–Satterthwaite degrees of freedom
p  = chance of a t this big if nothing changed (one-sided: "after is lower")

Improved            if p < 0.05
Got worse           if p > 0.95     (it got slower, and too much to be chance)
No clear change yet otherwise
Not enough data     if either side has fewer than 2 calls, or there is no variation at all
```

Worked example. Before = [10, 12, 14] and after = [6, 8, 10]. The means are 12 and 8. Each variance is 4, so t = 4 / √(4/3 + 4/3) = 2.449 with 4 degrees of freedom.

Steady. ERGent also checks whether the new level is **steady**. It draws the zone's daily chart after the action, with limits from the first 14 days after the action, and looks for a slow-side flag.

Watch out for. A better time after a fix does **not prove** the fix caused it. Other things may have changed. The check shows an *apparent* change. Staff judge the cause. If the after period is 14 days or shorter, the "steady" check judges those days against limits made from the same days.

12. Impact estimate

What it tells you. A best-case guess of how much better on-time performance could be if one stage were fixed. For example "on time 78% → 97%".

How to read it. It answers: "if this zone's slow stage were as fast as the city's typical call, how many more calls would be on time?" It is an **estimate, not a promise**.

The formula. Take the flagged stage (turnout or travel). Cap each call's stage time at the **city's median** for that stage (priority and last 14 days). Then recount.

```
new response = response − stage time + min( stage time, cap )
on time      = new response <= the call's response target
```

For the dispatch stage, "on time" means dispatch time at or under the processing target (64 s by default).

Worked example. Turnout [3, 1, 4, 1] minutes and travel [7, 7, 6, 9] minutes, target 9, city median turnout 1 minute. Responses are 10, 8, 10, 10, so 1 of 4 is on time (25%). With turnout capped at 1 minute they become 8, 8, 7, 10, so 3 of 4 are on time (75%).

Watch out for. Real results are usually smaller. A fix rarely brings every call down to the median.

13. Suggested actions for the supervisor (rules S1 to S6)

What they tell you. Fixed "if this, then suggest that" rules turn the numbers into something to try. Each suggestion shows its **evidence** and says "for review by qualified staff".

How to read them. The rules look at the **last 14 days** up to the as-of date. A stage's **share** is its excess divided by the zone's total excess (section 8). Rules S1 to S3 only apply to zones that are **amber or red**.

Rule	Tag	If	Then suggest
S1	Crew readiness	Turnout share is 50% or more	Review crew readiness at that zone's station. It names a shift when that shift holds 40% or more of the turnout excess **and** at least 1.5 times its share of the calls.
S2	Coverage	Travel share is 50% or more, 30% or more of late calls are coded "unit from another zone", and the zone borrows ambulances at least 1.5 times the city rate	Consider posting an extra ambulance in that zone during the busiest three hours of late calls
S3	Routes	Travel share is 50% or more, and "traffic or road closure" is the top code (at least 3 calls and at least 30% of late calls)	Check road works and alternative routes. It says "weekdays" if 80% or more of late calls fall on weekdays.
S4	Dispatcher cover	Citywide, all priorities: in an hour of the day with above-median call counts, the 90th percentile of dispatch time is over the processing target	Add dispatcher cover in those hours
S5	Maintenance	One ambulance has 2 or more breakdowns in the window	Schedule preventive maintenance for that ambulance
S6	Address checks	A zone has at least 5 late calls, and 20% or more of them have "unclear address"	Use an address-verification script

The formula for the evidence and the estimate is in sections 8 and 12.

Watch out for. The rules only see what the crews record. If delay reasons are missing, S2, S3, S5 and S6 cannot fire. The tool **decides nothing**. A qualified person decides.

14. Dispatcher tips (rules T1 to T6)

What they tell you. Live suggestions for the dispatcher, from the board right now and the last two shifts.

How to read them. They appear in **Suggested actions** on the History tab.

Tip	If	Then suggest
T1 Dispatch now	A call has waited longer than the processing target (64 s) with no dispatch	Dispatch it now. It names the nearest free ambulance: own zone first, then neighboring zones.
T2 Rebalance	A zone has ambulances but none free, and a neighbor has 2 or more free	Move one to stand by (from the neighbor with the most free)
T3 Cover	An ambulance is out of service	Ask the neighboring zones to cover
T4 Route	A zone has 3 or more calls and its **median** travel is at least 1.25 times the city's	Check the route. If 2 or more late calls share a delay reason, name it: reroute around traffic or road works, or keep an ambulance posted.
T5 Station alert	A zone has 3 or more calls and its **median** turnout is over the turnout target (60 s)	Alert that station's crews
T6 Dispatcher load	The city has 5 or more calls and the **90th percentile** of dispatch time is over 64 s	Ask the supervisor for a second dispatcher

Watch out for. Tips use small samples (a few calls), so they are prompts for your judgment, not verdicts. "Neighboring zones" means zones that touch on the map: A touches B and D; B touches A, C and E; C touches B and F; D touches A and E; E touches B, D and F; F touches C and E.

15. Slip-back

What it tells you. Whether a zone is drifting back to its old problem after a fix.

How to read it. A **Slip-back alert** names the zone and the warning sign, and the zone is at least amber on the map.

The formula. For each logged action, ERGent builds the zone's daily chart with limits from the **first 7 days after the action**. If a slow-side flag appears on a day **after those 7 days**, it raises an alert.

Watch out for. It needs enough days after the action. In the demo data, there is no slip-back for Zone D, which is the right answer.

Quick reference: the formulas on one page

Statistic	Formula
Response	`on_scene − dispatched`
SD	`sqrt( sum (x − mean)^2 / (n − 1) )`
Percentile (type 7)	`h = (n − 1) p`, then interpolate
Daily center and sigma	center = mean of daily means; `sigma_xbar` = SD of daily means
Daily limits	`center +/− 3 x sigma_xbar`
S chart	`B3 x s_bar` and `B4 x s_bar`
Every-call limits	`center +/− 3 x MR_bar / 1.128`
z	`(point − center) / sigma_xbar`
Zone box chart limits	`center +/− 3 x sigma_short / sqrt(n_zone)`
Ppk	`(ln USL − mean ln x) / (3 x SD ln x)`
Cpk	`(ln USL − mean ln x) / (3 x sigma_short)`
Expected on time	`100 x Phi(3 x Ppk)`
ANOVA F	`( SSB / (k − 1) ) / ( SSW / (N − k) )`
Welch t	`(mean_b − mean_a) / sqrt( s_b^2/n_b + s_a^2/n_a )`
Stage excess	`max(0, stage time − target)`

CHAPTER 5

Glossary and FAQ

Glossary

Terms are in alphabetical order.

Term	Meaning
90th percentile	The time that 9 in 10 calls beat. The slowest 1 in 10 calls take longer. The standard asks for it to be at or under the target.
Action	A fix that the supervisor logs, with a date, a zone, a stage, what was done and an owner. ERGent then checks whether it worked.
Alarm	A flag on the **slow** side of a control chart. A flag on the fast side only says "check the data".
ANOVA	A test that asks whether several groups (here, zones) really differ, or whether the gaps could be chance.
Average (mean)	The sum of the times divided by how many there are. It hides slow calls, so ERGent also shows percentiles.
Baseline	A stretch of normal days used to set a zone's normal range. In ERGent, the **chart period** plays this role.
CAD	Computer-aided dispatch. The city's system that keeps one row per 911 call and stamps the times.
Capability (Ppk, Cpk)	A measure of whether a stable process can meet its target reliably. ERGent shows it as a plain verdict.
Center line	The middle line of a control chart: the zone's normal level.
Chart period	How many days of history the cards use (14, 28 or 42). It also sets the control limits.
Chart signal	A strong warning on the daily chart: rule R1 fired, or two or more different slow-side rules fired in the last 7 days.
Clock speed	How fast the board clock runs when you log calls by hand (1x, 10x, 30x or 60x).
Common cause	Ordinary, everyday variation. The ups and downs inside the control limits.
Control chart	A chart of a process over time, with a center line and control limits. It shows when something unusual happens.
Control limits (UCL and LCL)	The upper and lower lines on a control chart, at 3 sigma from the center. They show what is **normal for the process**. They are not the target.

Term	Meaning
Crew turnout	The time from dispatch until the crew leaves (`dispatched` to `en_route`). Target: 60 seconds.
Delay reason	One of six codes the crew picks when a stage runs late: traffic or road closure, ambulance breakdown, unit from another zone, unclear address, weather, crew not ready.
Dispatch processing	The time from the call arriving until an ambulance is dispatched (`call_received` to `dispatched`). Target: 64 seconds.
False alarm	A flag that fires by chance when nothing has really changed. A few are expected.
Geometric mean	The "typical" value for skewed data: `exp` of the mean of the logs. ERGent uses it for a typical response time.
Log scale (ln)	A way of measuring that pulls in a long slow tail. ERGent's charts and tests use the natural log of the time.
Median	The middle value: half the calls are faster, half are slower.
Moving range (MR)	The size of the jump from one call to the next.
NFPA 1710	A US standard for fire and EMS response. It is the source of the priority 1 targets (64 s, 60 s and 480 s).
On time	A call whose response was at or under its target. The goal is at least 90% of calls.
p-value	The chance of seeing a difference this big if nothing had really changed. A small p (under 0.05) means the difference is probably real.
Percentile	The value that a given share of the data falls under. The 90th percentile is beaten by 90% of calls.
Poka-yoke	Mistake-proofing. The screen makes the wrong entry hard to make, for example by showing only the next valid button.
Priority	The urgency code for a call: 1 life-threatening, 2 urgent, 3 non-urgent. Each has its own response target.
Response time	The time from dispatch to arrival on scene (`dispatched` to `on_scene`). It is turnout plus travel. This is the prompt's metric.
Shift	An eight-hour block: day 06:00 to 14:00, evening 14:00 to 22:00, night 22:00 to 06:00.
Sigma (σ)	A measure of spread, like a standard deviation. Control limits sit 3 sigma from the center.
Slip-back	A zone drifting back to its old problem after a fix.
Special cause	Something that changed the process: a breakdown, road works, a new crew habit. A flag points to one.

Term	Meaning
Specification limit (spec limit)	The requirement, such as the 9-minute target. It comes from a standard, not from the process.
Stable	A process that stays inside its control limits with no flags. Capability only makes sense for a stable process.
Standard deviation (SD)	How spread out the values are around their mean. A big SD means less predictable service.
Subgroup	A small group of values charted as one point. In the daily chart, a subgroup is one zone-day. In the History chart, it is one zone.
Suggested action	A fixed "if this, then suggest that" rule on the data, shown with its evidence.
Synthetic data	Invented data, made by a script. ERGent contains no real patient data.
Travel time	The drive to the scene (`en_route` to `on_scene`).
Welch's t-test	A test of whether two groups have different averages that does not assume equal spread. ERGent uses it for before and after.
$\bar{X}$ chart (x-bar chart)	A control chart of group averages. ERGent's daily chart shows each day's average.
Zone	One of six areas of Ithaca (A to F). The outlines on the map are illustrative, not official EMS districts.

Frequently asked questions

1. Why not just use the monthly average?

An average blends fast and slow calls. A handful of very slow calls can hide inside a good-looking average. Take ten calls with times 4, 5, 6, 7, 8, 9, 10, 11, 12 and 20 minutes. The average is 9.2 minutes, close to a 9-minute target. But only 6 of the 10 were on time, and one patient waited 20 minutes. ERGent shows the 90th and 95th percentiles and on-time share next to the average, and splits each zone and each day. That is where the slow calls show up.

2. Why does ERGent use logs?

Response times are skewed. Most are quick, and a few are very slow. The log pulls in the slow tail, so control charts and tests behave. ERGent turns the results back into minutes before showing them, so you never have to think in logs.

3. What is the difference between amber and red?

On the Zone map, **amber ("Worth a look")** means one warning sign: either the daily chart shows a strong change, or the zone's 90th percentile is over its target. **Red ("Act now")** means both together. Amber is

worth checking. Red needs action. Some amber zones are false alarms, and the stricter amber rule makes them rare.

4. Why does Zone D's daily chart show no red dots when the Zone map is amber?

The control limits come from the **chart period** you picked. If a problem lasts for most of that period, it widens the limits and can hide itself on the daily chart. That is a trade-off the team chose for simplicity and consistency. The Zone map, Delay by stage and Capability cards do not hide it. Read the daily chart together with them.

5. What does "Can't judge yet" mean on the Capability card?

The zone's times changed during the period (it has a flag on its daily chart). A reliability score for a zone that is changing would mislead. Find and fix the cause first, then judge again.

6. What does "Not enough calls" mean?

A zone needs at least 30 calls in the chart period for a trustworthy score. With fewer calls, the numbers are too uncertain.

7. Is the data real?

No. All the data is synthetic. A script made it with a fixed random seed. It has no real patients, addresses or names. It is shaped like real field data so that the tool has something realistic to analyze. The Ithaca zone outlines are illustrative too.

8. Where do the targets come from?

For priority 1, from NFPA 1710, a US standard: 64 seconds to dispatch, 60 seconds for the crew to get moving and 8 minutes to travel, so 9 minutes from dispatch to arrival, each met by at least 90% of calls. **The priority 2 (15 minutes) and priority 3 (20 minutes) targets are our own assumptions**, because the standard does not set them. A supervisor can change every target with the **Targets** button.

9. Does ERGent dispatch ambulances or change staffing?

No. ERGent **flags and suggests**. Every suggestion is "for review by qualified staff", and the dispatcher's tips are "for the dispatcher's judgment". People make every decision.

10. What is the difference between a control limit and the target?

A **control limit** (UCL and LCL) comes from the process: it shows what is normal for a zone. The **target** (a specification limit) comes from a standard: it shows what is required. A zone can be within its normal range and still miss the target, or beat the target and still be changing. ERGent draws them as different lines so you never confuse them.